

Direct axis routing removes a cubic exponent loss

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Abstract

We audit a scheduling improvement to OpenAI’s integer multiplication manuscript. Its existing arbitrary-gap interchange primitive permits the CRT axis reversal and resampling axis exposure to use $O(d)$ field interchanges rather than $O(d^2)$ adjacent interchanges. The resulting layout cost removes the constraint that the dimension exponent must be as small as the network saving. Conditional on the upstream algorithmic interfaces, exact parameters give $T(n) = O(n(\log n)^{1-\kappa})$ with $\kappa = 2^{-78}$ using the previously audited $h = 46$ network. Keeping the original network and recurrence exponents instead gives $\kappa = 2^{-107}$. The earlier cubic ceiling concerns the original cost accounting and does not apply to the improved schedules.

1 Scope and primitive

The source is OpenAI’s *Integer multiplication below $n \log n$* , pinned at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

We use its fixed finite-alphabet Turing-machine model with a fixed finite number of one-dimensional tapes. The source algorithmic interfaces, including their uniform time bounds, remain assumptions of this conditional improvement.

Let V be the stored bit volume of a complete padded tensor with axes of lengths $t_i \in \{2^{\ell-1}, 2^\ell\}$ and complete coefficient records. Lemma `lem:chunk-swap` accepts the address shape

$$[P] \times [2^u] \times [G] \times [2^u] \times [B]$$

and interchanges the two width- u fields in $O(Vu^\tau)$ steps. Prefix, gap, and suffix lengths are arbitrary positive integers; B may include the entire coefficient word. The paragraph following the lemma already extends this to widths differing by one. Writing e for the extra bit gives

$$(e, x, G, y) \mapsto (e, y, G, x) \mapsto (y, G, e, x),$$

$$(x, G, e, y) \mapsto (e, x, G, y) \mapsto (e, y, G, x),$$

where x, y have equal width. Each case costs one interchange and one $O(V)$ single-bit move. Thus a nonadjacent full-field swap costs $O(V(1 + \ell^\tau))$. Empty equal-width pieces require no swap. These are permutations of complete records; they preserve every record’s internal bit order. No new network or interchange primitive is introduced.

*Prepared with assistance from OpenAI Codex. This is not independent review or endorsement by OpenAI.

2 The two schedules

Proposition 1 (Linear count of field interchanges). *Conditional on the stated stream and swap primitives, CRT axis reversal and exposure/restoration of every resampling axis cost $O(Vd(1+\ell^\tau))$, including descriptors and cleanup.*

Proof. For the CRT map, the padded input order is $(a_d, \dots, a_1)$. Swap positions j and $d+1-j$ for $1 \leq j \leq \lfloor d/2 \rfloor$. The disjoint swaps produce exactly $(a_1, \dots, a_d)$. In the existing triangular formula

$$b_i = a_i + \mu_i \sum_{j < i} a_j P_j \pmod{s_i}, \quad P_i = \prod_{j < i} s_j, \quad \mu_i = P_i^{-1} \pmod{s_i},$$

targets are still processed in order $d, d-1, \dots, 1$. All controls precede the target and still contain the required original digits. The inverse recovers digits in order $1, \dots, d$ and then reverses the field permutation. Padding insertion precedes the forward reversal; deletion follows the inverse. The original enumerations are therefore unchanged.

For resampling, swap the next named axis with the innermost field, process its lines, and undo that swap. An already innermost axis needs no swap. This uses at most $2(d-1)$ field interchanges per tensor map. Axis names, widths, and validity bounds move with their fields. A line fixes all other named coordinates and traverses the target in increasing numerical order. Their temporary physical order does not affect its validity predicate.

Keep the original axis-processing order and restore the original physical field order after every axis. The resulting operator is exactly the original coordinatewise map, with the same line arithmetic and truncations. Expansion writes all t_i outputs; compression writes s_i outputs and pads with zeros. The full box stays allocated. Invalid entries remain zero at the completed axis-map boundaries. An intermediate bit move or the internal arithmetic of a swap need not preserve an intermediate physical validity mask: only the completed permutation is used by the next numerical operation. Exact permutations preserve norms, so the contraction and error bounds are unchanged.

Each call collapses all intervening axes into its single spectator field. Since $t_i \geq 2$, $d \leq \log_2 T$, where $T = \prod_i t_i$. The named-axis list, record-width description, and all collapsed lengths have size polynomial in $\log(2V)$. A call descriptor can be computed by ordinary integer arithmetic in $\text{poly}(\log(2V)) = O(V)$ time. Thus list scans and updates are charged at every interchange. The primitives run sequentially, clean their workspaces, and reuse a fixed tape collection. All growing lists are stored data. Multiplying the per-interchange bound by the linear count proves the claim. $\square$

The d controlled CRT rotations cost $O(dV)$ as before. Numerical resampling costs and per-line setup do not change. In particular, no assertion that approximate maps commute is needed: the axis-processing order is preserved.

3 Changed accounting and explicit witnesses

The source has $p = \Theta(\log n)$, $V = \Theta(n)$, $d = \Theta(p^\epsilon)$ and $\ell = \Theta(p^{1-\epsilon})$. The old layout term and its exponent were

$$d^2(1 + \ell^\tau), \quad \tau + \epsilon(2 - \tau).$$

The proposition replaces these by

$$d(1 + \ell^\tau), \quad \tau + \epsilon(1 - \tau).$$

Consequently the fourth margin becomes

$$g_4 = (1 - \tau)(1 - \epsilon).$$

The other six margins retain their original formulas:

$$\begin{aligned} g_1 &= 1 - \epsilon(1 + c), & g_2 &= \epsilon c(1 - \tau), \\ g_3 &= \epsilon(1 - \lambda'), & g_5 &= \frac{1}{4} - \delta - \frac{3}{2}\epsilon, \\ g_6 &= 1 - \delta - \epsilon, & g_7 &= \epsilon. \end{aligned}$$

No analytic d^2 factor in the precision or prime-interval bounds is removed.

Proposition 2 (Parameter substitution). *For either $a = 2^{-50}$ with the original network, or $a = 9/500000000000$ with the previously audited $h = 46$ network, set*

$$\begin{aligned} \tau &= \sigma = 1 - a, & \beta &= \frac{1}{2}, & \delta &= \frac{1}{16}, & C_1 &= 20, \\ \epsilon &= \frac{1}{40}, & c &= a/2, & \lambda &= 1 - 3a^2/4, & \lambda' &= 1 - a^2/2. \end{aligned}$$

The modified assembly constraints and all recorded side conditions hold. The minimum final margin is $G = a^2/80$.

Proof. The layer inequalities follow from

$$\tau(1 + c/\beta) = (1 - a)(1 + a) = 1 - a^2 < \lambda < \lambda' < 1, \quad 1 - a/2 < \lambda'.$$

Also $\max\{\tau, \sigma\} < \lambda$. Both values of a are less than $1/16$, which is more than sufficient for these comparisons. The guard and dimension conditions are $\epsilon C_1 = 1/2 < 1$ and $\epsilon < 1/12$. Direct substitution gives

$$g_2 = g_3 = \frac{a^2}{80}, \quad g_4 = \frac{39a}{40}, \quad g_5 = \frac{3}{20}, \quad g_6 = \frac{73}{80}, \quad g_7 = \frac{1}{40}, \quad g_1 > \frac{1}{2}.$$

The first two are the minima for either stated a .

The remaining dependencies can be checked explicitly:

Quantity	Exponent or implication
Guard d^{20}	$p^{1/2} = o(p)$
Resampling α	$\Theta(p^{21/80}) = o(p^{1/2})$
Resampling γ	$O(p^{11/20}) = o(p)$
Prime-interval ratio	$p^{1-2\epsilon} = p^{19/20} \rightarrow \infty$
Chunk K	$\Theta(p^{a/80})$, hence $K/\log p \rightarrow \infty$
Axis width ℓ	$\Theta(p^{39/40})$, hence $K = o(\ell)$
Record length r	$2^{\Theta(p^{39/40})}$ dominates every fixed polynomial

Thus polynomially many exceptional repairs, descriptor preparations and per-line setup retain their asymptotic bounds. Since $d \rightarrow \infty$, prime search still takes $n^{o(1)}$ time. The explicit root calculation is the fixed-power comparison $d^{40} \leq b$. The original error bound still yields correct rounding beyond one common cutoff because $\gamma = o(b)$ and $p = 6b$. $\square$

Corollary 3 (Conditional multiplication bounds). *With the original network and recurrence exponents, the modified schedule permits $\kappa = 2^{-107}$. With the audited $h = 46$ network and rational recurrence saving, it permits $\kappa = 2^{-78}$.*

